Written Reflection

1. What was the hardest decision you made and why?

The hardest decision in this project was determining which outliers should be removed from the dataset. During exploratory data analysis, I identified a small number of houses with extremely large above-ground living areas *(GrLivArea)* but unusually low sale prices compared to similar properties. The challenge was deciding whether these observations represented genuine luxury properties, unusual market conditions, or anomalies that could distort the regression model.

I avoided removing all extreme values because large and expensive houses are realistic in housing markets and contain important information about high-end property pricing. Instead, I used a selective rule-based approach by investigating houses with very large living areas combined with relatively low sale prices. Only observations that appeared highly inconsistent with the overall market trend were removed. This decision balanced model stability with preservation of realistic housing variation and helped reduce the risk of over-cleaning the data.

2. What does your model get wrong? What do the rows with the biggest prediction errors have in common?

The model struggled most with unusual or luxury properties that could not be fully explained using numerical variables alone. Many of the largest prediction errors occurred in houses with uncommon feature combinations, such as very large homes, highly customized properties, or expensive houses with premium characteristics not captured by numerical measurements.

Because the assignment restricted the project to numerical features only, the model could not use important categorical information such as:

- neighbourhood quality,
- kitchen quality,
- exterior materials,
- architectural style,
- overall design appeal,
- or location desirability.

As a result, some homes with strong non-numerical advantages were either underpredicted or overpredicted by the model. The model also found it difficult to generalize for rare housing patterns that were not well represented in the training data.

3. If you had one more week, what would you try next?

If I had one more week, I would expand the project in several ways to improve predictive performance and model understanding.

First, I would include categorical variables using encoding techniques such as one-hot encoding or target encoding because many important housing characteristics are categorical in nature. This would likely improve prediction accuracy substantially.

Second, I would experiment with more advanced machine learning models such as:

- XGBoost,
- LightGBM,
- Random Forest Regression,

- and Elastic Net Regression.

I would also perform more extensive hyperparameter tuning and feature selection to identify the most important predictors.

Additionally, I would explore residual analysis and feature importance more deeply to better understand where the model performs poorly and how specific variables influence predictions.

Finally, I would improve the deployment side of the project by creating an interactive dashboard or simple web application that allows users to input property features and receive predicted house prices in real time.